Accessible Emergency Alert System

1. Introduction

In the modern world with climate change making disastrous natural events to occur frequently and with more force, benefits of spreading emergency alerts when they are needed is more crucial than ever. The need to have emergency alerts is well illustrated by the "Accessible Emergency Alert System Using Live Weather Forecast" project that intends to provide web-based alerts to concerned users on the prevailing weather conditions to enable them take appropriate measures on any adverse weather developments.

It provides a special for conveying all sorts of important notification and alert to the people with disability or sensory impairment such as the visual notification, the audible notification, and the haptic feedback. Using current live weather feed, the system can therefore give alerts on cold waves, areas of heavy rainfall, tsunamis and other unfavourable weather conditions which are peculiar to certain regions.

However, the ultimate goal of this project is to save peoples' lives by extending a Cs capabilities to alert the public via cheap technologies. By its intuitiveness, flexibility in settings and inaccessibility for the users the system enables them to timely inform about potentially dangerous weather conditions and protect themselves. It is also evidenced in this project how crucial accessibility is within emergency management furthermore, it also shows how the use of technologies to advance safer communities for everybody is achievable.

Thus the combined features such as alert options, preferences and real time data make the "Accessible emergency alert system" a unique means that closes the gap between weather information and the user action; or in other words, people with disabilities receive these critical alerts.

2. Brief Problem Statement

The "Accessible Emergency Alert System" is designed to be an emergency alert tool, which offers real-time weather-based emergency information to people residing in particular locations. The system will also inform the users about unfavourable weather conditions like freezing temperatures, rain, storm, etc and the users will avoid exposure to such weather conditions if necessary. To solve this problem, this solution meets the need of timely and accessible notifications to enhance safety and preparedness especially amongst vulnerable population.

3. Stakeholders

- **End Users:** The people at risk in extreme weather regions of their areas which encompasses the aged people, the disabled and families with youngsters.
- **Weather Data Providers:** Companies that offer real time weather information service (for instance Open Weather, AccuWeather etc).
- **Developers:** The individuals in charge of the management, development and maintenance of the alert system.
- **Local Authorities:** Owners of emergency services and local governments who are seeking for an application that will help them deliver alerts to the residents.
- **Accessibility Advocates:** Some organizations aiming on the issue and facilitation for the disabled people.

4. System Requirements

- **Integration with Weather APIs:** The system needs to retrieve weather information in real time from trustworthy API's.
- **Cross-Platform Availability:** The web-based application should however be able to be run from both a desktop and a mobile platform.
- **Alert Notification System:** The system needs to offer the notification in text form as well as voiced.
- **User Preference Customization:** A glimpse indicates that users should be able to choose their preferred forms of alert, their rate of occurrence and crisis standards.
- **Accessibility Compliance: (Optional)**The system has to be 100% friendly for disabilities and support such functionalities as screen readers and the like.

5. Functional Requirements

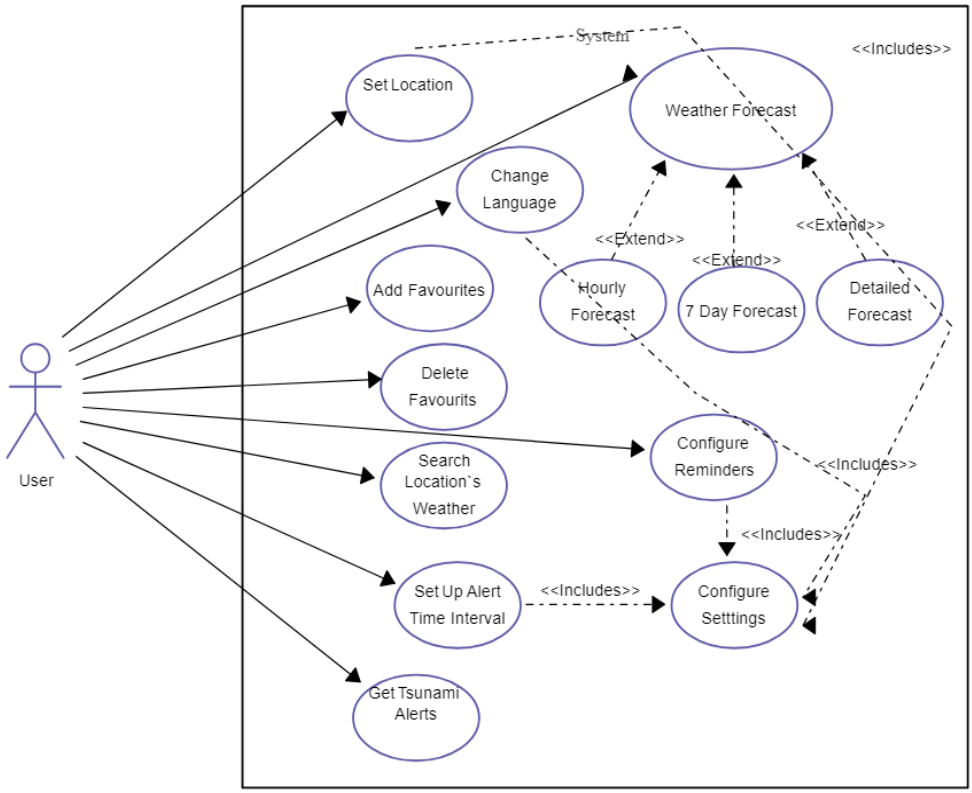
- **User Registration and Authentication:** End users should be able to register and login to enable them to set their alerts in line with their preferences.
- **Geolocation Detection:** Geolocation enable the system aspects to be able to identify users' current locations in order to provide weather based alerts.
- **Weather Data Fetching:** Use an API (e. g. Open Weather Map) to get current data on the weather.
- **Temperature Alert Notifications:** Activate additional text messages if the temperature is going down below certain level specified by the user.
- **Rain Alert Notifications:** Send intimation to the users that raining is going to occur through message and alarm alerts.
- **Custom Alert Scheduling:** Make it possible for the users to choose the specific time or frequency for the alerts for example the alert for cold may be required to be given in the two time of the day.

- **Auditory Alerts:** Trigger the tone to notify users about adverse weather conditions (a tune, for instance).
- **Visual Alerts:** Use 'pop-ups' on the website informing the visitors of the weather alert.
- **User Preference Management:** design an editable screen/field where users can change their alert settings for instance, the type and frequency of the alerts.
- **Weather Map Integration:** Display current real time maps to depict sensitive areas such as one that may depict a particular region as experiencing heavy rain.
- **Accessibility Features:** Make sure that all the text as a form of information is accessible to be read by screen readers and that all sounds have flashes.
- Alert History: Save the record of the alerts that has been sent to the user for other references.

6. Non-Functional Requirements

- **Performance:** Parallel users should be supported in the system without significant degradative impacts on the efficiency of the program.
- **Security:** The user's data or information such as location and preferences need to be protected through encryption.
- **Scalability:** Most important, system should have potential for adding new users and increase frequency of alarms.
- **Usability:** It is important that the main interface behind the application is user friendly, especially to those with computer literacy.
- **Accessibility:** WCAG 2 compliance of course is a limited aspect of dissability and accessibility compliance of Web sites on the Internet. 1 Working according to the AA standards for accessibility.
- **Reliability:** The availability of the system should be 99 % at least. 99% to make sure that the alert is on time.
- **Response Time:** These alerts should be given within the shortest possible time preferably not exceeding 30 seconds when the weather conditions are corresponding to those under study.

7. Use Case Diagram



8. Use Case Explanation

Set Location:

Enables the user to enter the location he or she desires to obtain weather information from, which the system relays to obtain the desired and accurate data.

Change Language:

Enables the user to switch the language in which the application's interface is displayed with the preferred language to the application.

Add Favourites:

Designed in such a way that it allows the user to select certain locations that he or she wants to have the option of setting as favourite locations for the purpose of receiving the weather forecast.

Delete Favourites:

Allows the user to 'unfavourite' specific place in their list of favourite places.

Search Location's Weather:

Enables the user to type in the location for which he or she is interested in the weather forecast for other than the current location.

Set Up Alert Time Interval:

Enables the user to set the frequency within which he or she desires to be given the weather alerts (e. g. hourly, daily).

Get Tsunami Alerts:

Alerts the user with Tsunami alerts for the chosen or the current location of a user or provides with the necessary warnings.

Configure Settings:

Is the extension that enables the user to adjust several application parameters, such as notification types, sounds and interface settings.

9. Individual Contribution

Name	**Student Id**	**Contribution**
Neeraj Reddy Aluka	11694856	Developed the alert system backend
Lekhitanand Bandi	11727419	Developed the alert system integrated APIs
Patel Sree Kohali	11694857	Designed the UI.
Neha Yarrapothu	11816807	Make all Diagram and Testing all the cases

10. Conclusion

Therefore, the project "Alert System Using Live Weather Forecast" effectively solves the worthwhile problem of emergency alerts in one's locality as well as in cases of natural calamities especially the targeted regions. This system dissects accurate live data of the weather into readily understandable notifications to enable the users, especially to take precautionary measures against the occurrence of events that are usually disastrous like the cold wave, heavy rainfall, or even a tsunami.

This project realizes its aim by using a web-based application that has features of sending alerts through the use of text, lights, and sounds. The system aims at being friendly to the users

by enabling those with different sensory impaired to receive and understand the alerts on time. Moreover, the alert feature enables the user to customize an alert, as well as the geographic area of interest and specific alert notifications according to an individual's requirements and environmental context.

Including features such as selecting the time interval for the alerts and availability of the application in several languages as well as clearly presented weather conditions, the project reveals rather effective measures toward increasing public security and literacy. The use case diagram depicts all the general flows, which are typical for the application, so that the user should be able to find the necessary information as soon as possible.

In summary, this project not only gives useful information to the people at the places of extreme weather occurrence but also reveal the applicability of the real-time data with the easy-available web-based interfaces to enhance the preparation and reaction. Possible future upgrades may include the addition of even more advanced analysis, the possibility to generate more alerts, and the inclusion of artificial intelligence and machine learning that will make the system's alert more accurate.